

FlowRes: Importance-Guided Endpoint Flow Matching for Speech Codec Restoration

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Abstract—Noise-robust neural speech codecs often suffer from perceptual degradation at low bitrates and low signal-to-noise ratios. While generative postfilters can improve quality, they typically overlook the varying distortion severity across time-frequency regions, limiting targeted restoration of perceptually critical regions. In this letter, we propose FlowRes, a restoration-importance-guided endpoint flow matching postfilter for noise-robust speech codecs. Specifically, a Restoration-Importance Predictor predicts an importance map to reweight the training objective, guiding the flow model to focus on regions requiring targeted restoration. Furthermore, we formulate post-decoding restoration as endpoint flow matching. By enabling direct prediction of the clean spectral target from intermediate states, this formulation naturally accommodates metric-oriented objectives and supports high-quality few-step restoration. Experimental results demonstrate that our method consistently improves the perceptual quality and intelligibility of codec-decoded speech, especially under challenging low-SNR conditions.

Index Terms—speech enhancement, speech coding, flow matching, postfiltering

I. INTRODUCTION

Noise-robust codecs [1]–[3] directly reconstruct clean speech from noisy inputs but still suffer substantial residual degradation at low signal-to-noise ratios (SNRs). To improve decoded quality, various codec-aware postfilters have been proposed, including GAN-based [4], adaptive-filtering [5], [6], and diffusion/flow-based approaches [7], [8]. While effective, they generally reconstruct the degraded speech as a whole without explicitly modeling region-dependent restoration demands.

This limitation is critical because joint speech enhancement and compression introduce spectrally non-uniform distortions, such as missing high-frequency details, localized spectral holes, and distorted low/mid-frequency formants [9]–[11]. Consequently, severely degraded regions may receive insufficient emphasis when optimized together with mildly affected ones. This motivates an importance-guided strategy that explicitly prioritizes regions with greater restoration needs.

Meanwhile, standard flow-based postfilters typically predict velocity fields and rely on iterative ODE integration, requiring multiple function evaluations for high-quality generation. Furthermore, this indirect parameterization makes the direct ap-

plication of metric-oriented perceptual objectives less straightforward. These issues motivate a clean-endpoint formulation that directly predicts the restoration target, providing a natural interface for metric-specific supervision while facilitating high-quality few-step inference.

We therefore propose **FlowRes**¹, an importance-guided endpoint flow matching postfilter for joint speech enhancement and compression. FlowRes combines (1) a Restoration-Importance Predictor (RIP) to reweight the objective according to region-wise restoration demand, and (2) an endpoint flow matching formulation that directly predicts the clean target, supporting metric-oriented optimization and high-quality few-step restoration.

II. PROPOSED METHOD

A. Overall Framework

The overall framework of the proposed FlowRes is illustrated in Fig. 1. Given a noisy waveform $\mathbf{x}_n \in \mathbb{R}^T$, we extract its complex spectrogram $\mathbf{S}_n \in \mathbb{C}^{F \times T'}$ via short-time Fourier transform (STFT). A *Noise-Robust Codec* processes $\mathbf{S}_n$ for joint enhancement and compression, yielding a relatively clean decoded spectrogram $\mathbf{S}_d$. Next, utilizing $\mathbf{S}_d$, decoder features, and encoder-side cues, a *Restoration-Importance Predictor* generates a binary map $\mathbf{M} \in \{0, 1\}^{F \times T'}$, where 1 indicates regions needing focused restoration and 0 denotes mild refinement. Finally, guided by $\mathbf{M}$, an *endpoint flow matching* postfilter iteratively refines $\mathbf{S}_d$ into a high-fidelity speech spectrogram (see Fig. 2 for detailed visualizations).

B. Noise-Robust Codec

We use FreqCodec [12] as the backbone. The encoder extracts latents via four 2-D residual convolutional blocks, discretized by a finite scalar quantizer (FSQ) [13] to ensure stable training without learnable codebooks or commitment losses. A mirrored decoder then reconstructs these into $\mathbf{S}_d$. To achieve **joint compression and enhancement**, the codec is trained on paired noisy-clean speech.

C. Restoration-Importance Predictor

Inspired by the concept of importance maps in image compression [14]–[17] and neural speech coding [18], the Restoration-Importance Predictor (RIP) identifies time-frequency regions requiring targeted restoration. It takes three inputs: the decoded spectrogram $\mathbf{S}_d$, decoder features $\mathbf{F}_d$, and an encoder-side cue $\mathbf{Z}_s$. Specifically, the cue branch extracts

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¹A demo page is available at <https://ukitenzai.github.io/FlowRes.demopage>.

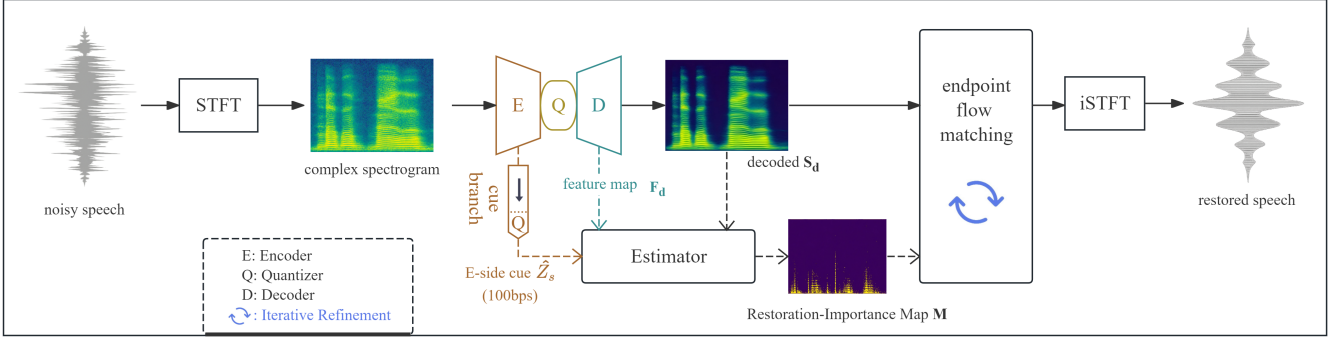


Fig. 1. Overall framework of the proposed **FlowRes**. The **circular blue** denotes **iterative refinement at inference time**, where the endpoint flow matching postfilter enables efficient few-step restoration.

compact pre-quantization information from encoder features $\mathbf{F}_e$ to provide source-aware cues that facilitate the prediction of $\mathbf{M}$. This representation is discretized via FSQ as $\hat{\mathbf{Z}}_s$ and transmitted at 100 bps.

The budget of 100 bps is empirically determined by measuring the additional information from the encoder. Specifically, we compute the binary difference $\Delta\mathbf{M} = \mathbf{M}_0 \oplus \mathbf{M}_{\text{full}}$ between the decoder-only importance map ($\mathbf{M}_0$) and the map utilizing unquantized encoder cues ($\mathbf{M}_{\text{full}}$). Compressing $\Delta\mathbf{M}$ with JBIG2² yields an average rate of 91.3 bps. Since this 100 bps budget is sufficient to capture the essential encoder-side cues, we apply this fixed side-information allocation across all total codec bitrates.

The inputs are projected by embedding functions $\phi(\cdot)$ and fused via channel-wise concatenation into a representation $\mathbf{H} = [\phi_{\text{spec}}(\mathbf{S}_d), \phi_{\text{feat}}(\mathbf{F}_d), \phi_{\text{side}}(\hat{\mathbf{Z}}_s)]$. An NCSN++ [19] backbone processes $\mathbf{H}$ to predict a normalized continuous map $\tilde{\mathbf{M}} \in (0, 1)^{F \times T'}$ via a sigmoid function, which is then binarized into $\mathbf{M} \in \{0, 1\}^{F \times T'}$ using a threshold of 0.5. Here, $M_{f,t} = 1$ indicates regions needing focused restoration.

During the joint training of the RIP and the cue branch, we utilize paired clean and noisy spectrograms ($\mathbf{S}_c, \mathbf{S}_n$). The noisy input $\mathbf{S}_n$ is first processed by the Noise-Robust Codec to yield the decoded spectrogram $\mathbf{S}_d$. Since the binarization of the importance map is non-differentiable, we apply the straight-through estimator (STE) during back-propagation:

$$\mathbf{M}_{\text{STE}} = \text{stopgrad}(\mathbf{M} - \tilde{\mathbf{M}}) + \tilde{\mathbf{M}}. \quad (1)$$

Subsequently, a hybrid magnitude spectrogram $|\mathbf{S}_h|$ is constructed by combining the codec-degraded representation $|\mathbf{S}_d|$ and the clean target $|\mathbf{S}_c|$ according to the estimated importance map:

$$|\mathbf{S}_h| = (1 - \mathbf{M}_{\text{STE}}) \odot |\mathbf{S}_d| + \mathbf{M}_{\text{STE}} \odot |\mathbf{S}_c|, \quad (2)$$

where $\odot$ denotes element-wise multiplication.

Following adversarially learned attention mechanisms in image translation [20], [21], we use the discriminator as a learned criterion to provide task-driven supervision for the restoration-importance map. The adversarial objective encourages the RIP to emphasize regions whose replacement most effectively reduces the discrepancy between the hybrid and

clean spectra, while the sparsity constraint prevents the trivial solution of selecting the entire spectrogram.

$$\mathcal{L}_D = \mathbb{E} \left[(1 - D_\eta(|\mathbf{S}_c|))^2 \right] + \mathbb{E} \left[(D_\eta(|\mathbf{S}_h|))^2 \right], \quad (3)$$

$$\mathcal{L}_{\text{RIP}} = \mathbb{E} \left[(1 - D_\eta(|\mathbf{S}_h|))^2 \right] + \lambda_{\text{sp}} \text{mean}(\tilde{\mathbf{M}}), \quad (4)$$

where λ_{sp} controls the sparsity strength. Driven by this objective, the cue branch implicitly learns to transmit task-oriented pre-quantization information without explicit supervision.

D. Endpoint Flow Matching Postfilter

Inspired by FlowDec [8], we employ a flow matching postfilter to restore decoded speech. To support high-quality few-step restoration and metric-oriented optimization, we adopt an endpoint-supervised formulation [22], which directly predicts the clean target from intermediate states. Specifically, the network g_θ is trained to directly predict the clean endpoint $\mathbf{S}_1 = \mathbf{S}_c$ from an intermediate state $\mathbf{S}_t$:

$$\hat{\mathbf{S}}_1 = g_\theta(\mathbf{S}_t, t \mid \mathbf{c}, \mathbf{M}). \quad (5)$$

This formulation offers two key advantages: it enables the direct application of perceptual objectives to the output, and it facilitates high-quality few-step generation with a reduced number of function evaluations (NFEs). A formal analysis is provided in the supplementary material.

In the forward process, the initial state $\mathbf{S}_0$ is sampled from a conditional Gaussian centered at the decoded speech $\mathbf{S}_d$, which also serves as the condition $\mathbf{c} = \mathbf{S}_d$. The intermediate state $\mathbf{S}_t$ is obtained via linear interpolation:

$$\mathbf{S}_0 = \mathbf{S}_d + \epsilon, \quad \mathbf{S}_t = (1 - t)\mathbf{S}_0 + t\mathbf{S}_1, \quad (6)$$

where ϵ is a complex Gaussian noise matrix with i.i.d. elements drawn from $\mathcal{CN}(0, \sigma^2)$, and $t \sim \mathcal{U}(0, 1)$.

To focus the postfilter on critical regions, we reweight the flow matching objective using the weighting map $\mathbf{M} + 1$, which assigns a weight of 2 to distorted regions ($M_{f,t} = 1$) and 1 otherwise ($M_{f,t} = 0$). The restoration-importance-weighted endpoint flow matching loss is formulated as:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, \mathbf{S}_0, \mathbf{S}_1} \left[\text{mean} \left((\mathbf{M} + 1) \odot |\hat{\mathbf{S}}_1 - \mathbf{S}_1|^2 \right) \right]. \quad (7)$$

Here, $|\cdot|^2$ denotes the squared magnitude of the complex error, yielding an $F \times T'$ time-frequency error map that matches the

²<https://github.com/agl/jbig2enc>

TABLE I
MODEL, BITRATE, AND TRAINING CONFIGURATIONS

Configuration	Setting
Sampling rate	16 kHz
STFT (FFT/win/hop)	512 / 480 / 160
Encoder	4×128-ch., temporal downsampling (2, 1, 1, 1) Output projected to 4 frequency bins and then flattened
Main latent	50 Hz
Cue branch	Additional ×5 temporal downsampling
Cue latent	10 Hz
1 kbps main stream	900 bps, $\mathbf{L} = [8, 8, 8, 8, 8, 8]$
1 kbps cue stream	100 bps, $\mathbf{L} = [4, 4, 4, 4, 4]$
6 kbps main stream	5900 bps, $\mathbf{L} = \{[8, 8, 8, 8, 8, 8] \times 6, [4, 4, 4, 4, 4]\}$
6 kbps cue stream	100 bps, $\mathbf{L} = [4, 4, 4, 4, 4]$
Training iterations	Codec/RIP/Postfilter: 250k/50k/50k
$\lambda_{\text{sp}} / \lambda_{\text{PESQ}}$	10 / 1
σ / δ	0.487 / 0.03
ODE solver	Euler

dimensions of $\mathbf{M}$.

Furthermore, to impose perceptually motivated constraints, we apply a differentiable PESQ surrogate loss³ between the time-domain output $\hat{\mathbf{x}}_1$ (obtained via ISTFT) and the clean reference $\mathbf{x}_c$. Minimizing this surrogate loss improves PESQ-aligned perceptual quality. The overall training objective is:

$$\mathcal{L}_{\text{post}} = \mathcal{L}_{\text{FM}} + \lambda_{\text{PESQ}} \mathcal{L}_{\text{PESQ}}(\hat{\mathbf{x}}_1, \mathbf{x}_c), \quad (8)$$

where λ_{PESQ} is a balancing hyperparameter.

During inference, we generate the restored speech by integrating the learned deterministic flow starting from $\mathbf{S}_0$. Since the network predicts the clean endpoint, the velocity field is derived as

$$\mathbf{v}_\theta = (\hat{\mathbf{S}}_1 - \mathbf{S}_t) / (1 - t). \quad (9)$$

The final restored spectrogram is obtained as:

$$\hat{\mathbf{S}} = \mathbf{S}_0 + \int_0^1 \frac{g_\theta(\mathbf{S}_t, t \mid \mathbf{c}, \mathbf{M}) - \mathbf{S}_t}{1 - t} dt. \quad (10)$$

In practice, to avoid numerical singularities at the boundaries (due to the log-scaled time embedding at $t = 0$ and the velocity denominator at $t = 1$), the integration is performed over a truncated interval $[\delta, 1 - \delta]$ with a small constant $\delta > 0$.

III. EXPERIMENT

A. Model Configuration

The noise-robust FreqCodec is trained with noisy speech as input and the corresponding clean speech as target, thereby performing joint speech enhancement and compression. The main model and training configurations are summarized in Table I.

The encoder-side cue branch takes the channel-wise concatenation of the output feature maps from all encoder blocks and further downsamples the temporal dimension by a factor of 5.

B. Datasets

We use LibriTTS [23], VCTK [24], and AISHELL-3 [25] as clean speech data, and VoiceBank+DEMAND [26] and DNS-Challenge [27] as paired noisy-clean data. All audio is resampled to 16 kHz, and the noisy evaluation set is restricted to SNRs from −5 to 5 dB.

C. Training and Evaluation Settings

Training and Inference Settings: The codec, RIP with the cue branch, and endpoint flow matching postfilter are trained sequentially for 250k, 50k, and 50k iterations, respectively, with preceding modules frozen at each subsequent stage. During inference, Euler integration is performed over $[\delta, 1 - \delta]$.

Evaluation: Objective evaluation uses 200 noisy-clean pairs. We report PESQ [28], SCOREQ [29], and UTMOS [30] for speech quality, together with word error rate (WER) computed by Whisper-tiny [31] for intelligibility. For subjective evaluation, 10 listeners (5 females and 5 males, aged 22–29) conducted MUSHRA tests [32] on 20 clean and 20 noisy utterances. The corresponding uncompressed clean speech served as both the explicit and hidden reference, while its 3.5-kHz low-pass-filtered version served as the hidden anchor. Every listener evaluated all selected utterances, and mean scores with 95% confidence intervals are reported. All participants provided informed consent.

Baselines: We compare the proposed method with four representative baselines:

- **FreqCodec** [12]: the underlying codec without postfiltering.
- **FreqCodec + One-step NCSN++** [19]: an NCSN++-based one-step postfilter.
- **FreqCodec + ScoreDec-style Postfilter** [7]: an adapted score-based diffusion postfilter conditioned on the FreqCodec output.
- **FreqCodec + FlowDec-style Postfilter** [8]: an adapted conditional flow matching postfilter following FlowDec.

Note that while all baselines operate at a total bitrate of 1 kbps or 6 kbps, our proposed FlowRes explicitly allocates this identical budget as 900+100 bps or 5900+100 bps, as summarized in Table I.

Computational Complexity: For a capacity-controlled comparison, all postfilters are trained on the same data and use NCSN++ as the main backbone. The baseline backbones are slightly enlarged to approximately 27.1M parameters, while the complete FlowRes, including the RIP and cue branch, contains 26.8M parameters, ensuring comparable model capacity. Real time factor (RTF) is measured on a single NVIDIA RTX 4090 under the same implementation environment.

TABLE II
OBJECTIVE EVALUATION RESULTS UNDER THE NOISY CONDITION

Model	Bitrate (bps)	PESQ↑	WER(%)↓	SCOREQ↓	UTMOS↑
FreqCodec	1k	1.60	55.5	0.81	2.24
NCSN++	1k	1.64	55.3	0.78	2.30
ScoreDec	1k	1.76	52.7	0.61	2.47
FlowDec	1k	1.82	49.3	0.52	2.58
FlowRes	1k	1.92	39.2	0.50	2.66
FreqCodec	6k	2.05	41.4	0.47	2.73
NCSN++	6k	2.09	39.9	0.44	2.79
ScoreDec	6k	2.13	37.1	0.38	2.86
FlowDec	6k	2.19	36.2	0.33	2.98
FlowRes	6k	2.35	30.8	0.30	3.03

³<https://github.com/audiolabs/torch-pesq>

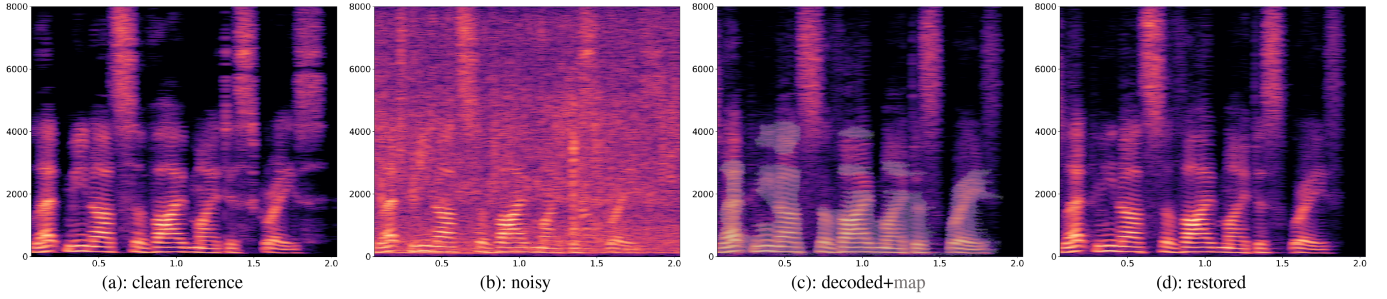


Fig. 2. Speech spectrograms before and after coding and restoration. The restoration-importance map in (c) is visualized by overlaying a semi-transparent mask on the decoded spectrogram.

TABLE III
COMPLEXITY AND INFERENCE EFFICIENCY OF DIFFERENT
POST-FILTERING METHODS

Model	Params (M)↓	NFE↓	RTF↓
NCSN++ (non-iterative)	27.1	—	0.017
ScoreDec	27.1	50	0.860
FlowDec	27.1	6	0.130
FlowRes	26.8	3	0.053

TABLE IV
ABLATION RESULTS UNDER THE NOISY CONDITION AT 6 KBPS

Model	PESQ↑	WER(%)↓	SCOREQ↓	UTMOS↑
FlowRes@6kbps (NFE=3)	2.35	30.8	0.30	3.03
w/o RIP	2.29	32.7	0.35	3.01
w/o $\mathcal{L}_{\text{PESQ}}$	2.23	30.9	0.32	3.06
– w/o endpoint supervision	2.12	40.5	0.41	2.79

D. Results Analysis

As shown in Table II and Fig. 3, FlowRes achieves the best overall objective performance under noisy conditions, while the MUSHRA results further demonstrate subjective improvements under both clean and noisy conditions. Beyond the targeted PESQ gains, the improvements in WER demonstrate enhanced intelligibility, while SCOREQ, UTMOS, and MUSHRA provide complementary evidence of improved perceptual quality. Together, Tables II and III demonstrate high-quality few-step inference with favorable computational efficiency, while the restoration-importance guidance improves restoration fidelity.

Furthermore, Fig. 2 qualitatively illustrates the region-dependent distortions discussed above. Compared with the clean reference in Fig. 2(a), the decoded spectrum shows more evident degradation around several formant-rich low-frequency and high-frequency regions, while the predicted importance map in Fig. 2(c) assigns higher weights to many of these areas. Guided by this map, the flow-matching postfilter emphasizes regions with greater restoration needs and recovers clearer spectral structures, as shown in Fig. 2(d).

E. Ablation Study

Table IV presents the ablation study on FlowRes’s key components.

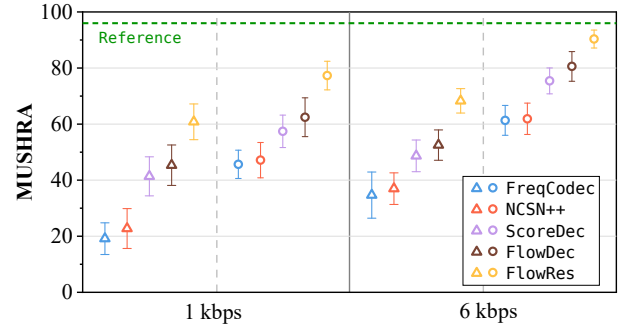


Fig. 3. Subjective MUSHRA scores for noisy and clean speech. $\triangle$ denotes noisy speech, while $\circ$ denotes clean speech. Colors indicate different methods: FreqCodec, NCSN++, ScoreDec, FlowDec and FlowRes.

First, removing the RIP and reallocating its 100 bps side-information budget to the main codec stream degrades reference-based metrics (PESQ, WER, and SCOREQ) but leaves the no-reference UTMOS largely unchanged. This suggests that the RIP improves restoration fidelity to the clean reference by prioritizing regions with greater restoration needs, while having a relatively limited effect on no-reference naturalness.

Next, removing $\mathcal{L}_{\text{PESQ}}$ leads to a noticeable PESQ drop of about 0.1, whereas the remaining metrics show no pronounced changes. This confirms that the metric-specific loss successfully provides targeted PESQ enhancements.

Finally, removing endpoint supervision on top of $\mathcal{L}_{\text{PESQ}}$ replaces the proposed endpoint objective with a conventional FlowDec-style flow-matching formulation. At NFE=3, this degraded model performs worse than FlowDec (NFE=6) in Table II. These results support the benefit of endpoint supervision for maintaining restoration quality under few-step inference.

IV. CONCLUSION

We propose FlowRes, an importance-guided endpoint flow matching postfilter for noise-robust neural speech codecs. FlowRes combines a Restoration-Importance Predictor for prioritizing critical time-frequency regions with endpoint supervision for metric-oriented, few-step restoration. Experiments demonstrate improved perceptual quality and intelligibility, particularly under challenging low-SNR conditions.

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